

Differentiable E-Z Reader: A Trainable Cognitive Cascade for Word-Level Eye-Tracking Prediction

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Abstract

Cognitive cascade models of reading such as E-Z Reader are interpretable but hand-tuned, while neural reading-time predictors are accurate but cognitively opaque. We bridge this gap with the first end-to-end-trainable instantiation of an E-Z Reader-style cascade with a transformer language-model backbone, jointly predicting word-level first-fixation duration (FFD), gaze duration, total reading time (TRT), and skip rate from raw text. A dual-context (dualctx) specialization resolves a structural FFD/skip trade-off, supported by lesion and cross-prediction evidence. The cascade is competitive with strong baselines on time metrics. Most strikingly, when cascade scalars are refit by reading-style group, the model recovers a textbook E-Z Reader prediction without supervision: *slow readers differ from fast readers in lexical-stage parameters but not in motor or skip-decision parameters*.

Keywords: eye-tracking; cognitive modelling; E-Z Reader; language models; reading; interpretability

Introduction

Eye-tracking provides a moment-to-moment record of reading. Predicting where readers fixate, for how long, and which words they skip is foundational to psycholinguistic theory and underpins applications ranging from reading-difficulty diagnosis and dyslexia screening to assistive interfaces and adaptive educational technology. Two largely disjoint research programs have produced predictive models, and neither has both the cognitive interpretability and the trainability that practitioners need.

The first program, in cognitive psychology, builds explicit cascade models. Words pass through a fixed sequence of named processing stages—familiarity check (L_1), lexical access (L_2), labile and non-labile saccade programming (M_1, M_2), and eye execution—each with a labelled parameter. E-Z Reader (Reichle, Rayner, & Pollatsek, 2003; Reichle, 2009), SWIFT (Engbert, Nuthmann, Richter, & Kliegl, 2005) and OB1-Reader (Snell, van Leipsig, Grainger, & Meeter, 2018) are prominent examples. These models are theoretically grounded and interpretable, but their parameters are hand-tuned to behavioural data and they require pre-computed predictability (e.g. cloze probabilities). They are simulators rather than differentiable functions of raw text.

The second program, in NLP, treats eye-tracking as a sequence-prediction task. Linear regressions on language-model surprisal (Smith & Levy, 2013; Goodkind & Bicknell, 2018), neural attention models such as NEAT (Hahn & Keller, 2016), and the CMCL 2021 shared-task winners

(Hollenstein, Pirovano, Zhang, Jäger, & Beinborn, 2021; Oh & Fazeli, 2021) all train end-to-end on raw text and obtain high predictive correlations. None preserve a cognitive cascade or expose interpretable psycholinguistic parameters.

Research question. Can a cognitive reading cascade be made fully differentiable and trained end-to-end against word-level eye-tracking targets without losing the interpretability of its named stages?

We decompose this into three sub-questions and corresponding hypotheses, each tied to a distinct strand of theory:

- SQ1.** Does a differentiable cascade match neural baselines on word-level eye-tracking prediction? **H1:** An E-Z Reader cascade with a transformer-LM front-end matches strong neural baselines on time-metric prediction (FFD, gaze, TRT).
- SQ2.** Do FFD and skip prediction require dissociable contextual representations? **H2:** A single shared context head will trade off between FFD and skip; a dualctx architecture will dissociate them, evidenced by lesion-based double dissociation.
- SQ3.** Does the trained cascade recover known psycholinguistic predictions about individual differences? **H3:** Following the prediction of Reichle (2009), refitting cascade scalars by reading-speed group will localize differences to lexical-stage parameters ($\delta, \epsilon, \lambda_{\text{refix}}$) and not to motor (M_1, M_2) or skip-decision (τ) parameters.

Methods

Data. We train on the GECO corpus (Cop, Dirix, Drieghe, & Duyck, 2017) (English monolingual subset, 5,285 sentences, 54,362 word tokens, 14 readers), splitting by text into 30,601 training, 4,494 validation, and 5,501 test word-aggregated sentences. We report cross-corpus generalization on Provo (Luke & Christianson, 2018) (2,654 word-aggregated sentences). All targets are aggregated across readers per word.

Model. A TinyLlama-1.1B (Zhang, Zeng, Wang, & Lu, 2024) language model produces per-token hidden states. The lower 75% of layers are frozen; only the upper layers are fine-tuned. A learnable projection maps token states to a

256-d per-word vector. Two specialized contextual heads— $\text{ctx-head}_{\text{FFD}}$ and $\text{ctx-head}_{\text{skip}}$ —produce two scalar context corrections per word. These feed a differentiable E-Z Reader cascade with named cog scalars (Figure 1).

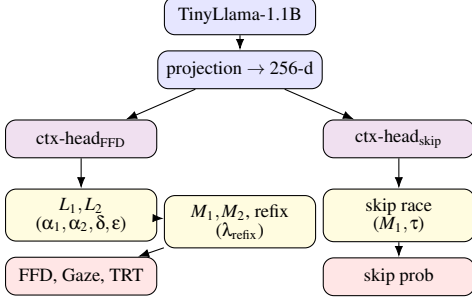


Figure 1: Architecture. Transformer LM features (blue) feed two specialized ctx-heads (purple); a differentiable E-Z Reader cascade with named cog scalars (yellow) integrates them and produces four eye-tracking measures.

Cascade equations. Let w be the current word with length ℓ_w and normalized log-frequency f_w , and let c_w^p denote the ctx-head output for path $p \in \{\text{FFD}, \text{skip}\}$. Familiarity-check time follows the Reichle (2003) frequency form, with eccentricity correction: $L_{1,w}^p = (\alpha_1 + \alpha_2 f_w + c_w^p) \cdot \epsilon^{(\ell_w-1)/2}$. Lexical access is $L_{2,w} = \delta \cdot L_{1,w}^{\text{FFD}}$. The cascade then computes $\text{FFD}_w = L_{1,w}^{\text{FFD}} + M_1 + M_2$, $\text{Gaze}_w = \text{FFD}_w + P_{\text{refix}}(\ell_w) \cdot (L_{2,w} + M_1 + M_2)$, and $\text{TRT}_w = \text{Gaze}_w + I$, with $I = M_2$ tied. Refixation probability is parameterized by λ_{refix} and a learned pivot. The skip path computes $P(\text{skip}_w) = \sigma((M_1 - L_{1,\text{next}}^{\text{skip}})/\tau)$ with parafoveal preview baked into the eccentricity term.

Training. Cog scalars are initialized at Reichle (2003) values and learned by backprop through the cascade. We use separate learning rates for the LM (2×10^{-5}), heads (5×10^{-4}), and cog scalars (3×10^{-4}); 5 epochs; 5 random seeds. Loss is the unweighted sum of MSE on FFD, Gaze, TRT and BCE on skip.

Why dualctx? A single shared head must encode both within-word lexical difficulty (for FFD/Gaze/TRT) and next-word parafoveal-preview ease (for skip). These are different functions of context. Splitting the head into two paths breaks the trade-off; the consequences are tested empirically below.

Baselines. Five baselines span the predictive-modelling space: (i) linear regression on log-frequency, predictability and length; (ii) LightGBM with engineered features and GPT-2 surprisal; (iii) GPT-2 surprisal in linear regression; (iv) BERT regression (Devlin, Chang, Lee, & Toutanova, 2019); (v) the CMCL 2021 RoBERTa winner (Oh & Fazeli, 2021). Comparison is on the same GECO test split and on Provo. Code, splits and trained checkpoints are released for full reproducibility (see Code statement).

Results

Predictive performance (H1)

Table 1 reports per-corpus Pearson r on the four metrics. Our cascade is competitive with neural baselines on time metrics on both corpora and lags flat regressors on skip prediction.

Table 1: Pearson r between predicted and human-aggregated values on GECO test and Provo. “5s” = mean over 5 seeds (std < 0.03). Bold: best per column.

Model	GECO				Provo			
	TRT	FFD	Gaze	skip	TRT	FFD	Gaze	skip
linear	.43	.18	.36	.72	.62	.32	.61	.85
GPT-2 surp.	.43	.18	.37	.62	.65	.32	.63	.76
LightGBM	.47	.23	.41	.74	.65	.33	.63	.88
BERT (5s)	.45	.22	.39	.64	.60	.30	.57	.77
OSU RoBERTa (5s)	.42	.14	.34	.74	.59	.25	.59	.87
Ours (5s)	.43	.19	.38	.39	.60	.26	.60	.54

On time metrics our cascade is within ~ 0.03 of the strongest single baseline (LightGBM, which uses an engineered feature set). On skip prediction it lags by 0.2–0.4 in correlation. We treat the skip gap as a structural cost of the cascade (§Discussion). H1 is therefore partially supported: the cascade matches baselines on time metrics but does not on skip.

Dualctx dissociation (H2)

Two analyses test the dualctx prediction. *Lesion study:* zeroing $\text{ctx-head}_{\text{FFD}}$ harms only time metrics ($\Delta r \approx -0.04$ on TRT/FFD/Gaze, $\Delta r_{\text{skip}} = 0$); zeroing $\text{ctx-head}_{\text{skip}}$ harms only skip ($\Delta r_{\text{skip}} = -0.18$, time metrics unchanged). Replacing the cog parameters with constants destroys time metrics ($\Delta r_{\text{TRT}} = -0.17$ for shuffled L_1) and skip ($\Delta r_{\text{skip}} = -0.37$). *Cross-prediction:* controlling for log-frequency and word length, the FFD-head’s per-word output predicts human TRT, FFD and Gaze (partial $r = +0.18, +0.15, +0.17$) but not skip (-0.10); the skip-head’s output predicts skip ($+0.09$) but not time metrics (~ 0). Both analyses show a clean double dissociation, supporting H2.

Surprisal decomposition

To check whether the contextual heads carry information beyond raw LM surprisal, we ran a hierarchical regression on GECO test (5,501 words). The cascade’s L_1 correlates with TinyLlama surprisal at $r = 0.35$. Controlling for surprisal, log-frequency, and length, L_1 retains a partial correlation with human TRT of 0.18. L_1 adds $\Delta R^2 = 0.027$ over surprisal-and-controls, while surprisal adds only $\Delta R^2 = 0.003$ over L_1 -and-controls. Replacing $\text{ctx-head}_{\text{FFD}}$ with a learned scalar times raw surprisal degrades all metrics on GECO and the skip degradation *grows* cross-corpus on Provo ($\Delta r_{\text{skip}} = -0.25$). The contextual heads encode information beyond surprisal that is robust across corpora.

Per-participant individual differences (H3)

We split the 14 GECO readers at the median mean-RT (163 ms) into a fast group ($n=7$, mean RT=129 ms) and a slow group ($n=7$, mean RT=194 ms). For each group we fine-tune *only the cascade’s nine cog scalars* on that group’s pooled data, freezing the LM and ctx-heads. Figure 2 shows the result.

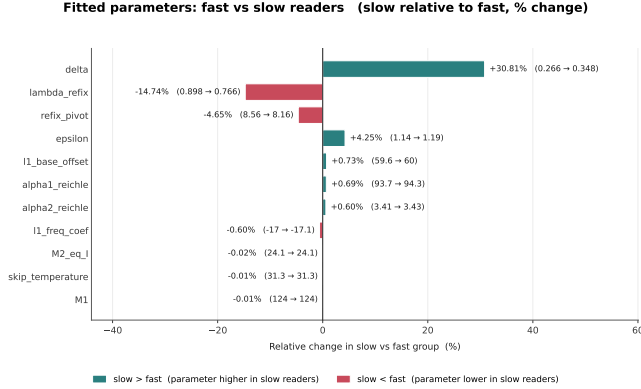


Figure 2: Cog scalars refit per reading-style group. Lexical-stage parameters ($\delta, \lambda_{\text{refix}}, \epsilon, \text{refix_pivot}$) shift substantially between fast and slow readers. Motor parameters (M_1, M_2) and skip-decision sensitivity (τ) are essentially invariant ($|\Delta| < 0.05\%$).

The fits reveal a clean dissociation: δ rises by +30.8% from fast to slow ($0.266 \rightarrow 0.348$), λ_{refix} drops by -14.7%, ϵ rises by +4.2%; M_1, M_2 change by $< 0.05\%$ (M_1 : $123.71 \rightarrow 123.69$), and τ is unchanged ($31.34 \rightarrow 31.34$). H3 is supported: reading-speed differences localize to lexical-stage parameters, exactly as Reichle (2009) predicts. We did not engineer this outcome; the cascade structure plus group-level fits produced it from optimization alone.

Discussion

The three hypotheses receive differentiated support. **H1** holds for time metrics but not for skip—the cascade lags flat regressors badly on skip rate. We attribute this to a structural constraint: the cascade computes $P(\text{skip})$ as a race that must share M_1 and L_1 with the time-metric path, so it cannot fit skip independently of how it fits FFD/Gaze/TRT. Flat regressors face no such constraint. This is the architecture working as designed; the gap is the price of cognitive plausibility. **H2** is supported by both lesion and cross-prediction evidence (§Results). **H3**—the recovered Reichle 2009 prediction—is the strongest cog-plausibility evidence in this work.

Weaknesses, ranked by impact on the central claim. (1) *The per-group dissociation rests on $n=7$ readers per group with no bootstrap CIs*; the headline finding therefore needs uncertainty quantification before it can be treated as established, not merely suggestive. (2) *The skip-prediction gap is large* (0.39 vs. 0.74 on GECO); a residual analysis by word

length and frequency would diagnose whether this is calibration error or a structural blind spot. (3) *The L_2 stage is degenerate* ($L_2 = \delta L_1$), so L_2 -only lesions are uninformative; a faithful L_2 requires an additional contextual signal absent here. (4) *Cross-corpus generalization on TRT and FFD is weaker than on Gaze and skip*, suggesting GECO-specific tuning that does not transfer. (5) The cascade computes per-word *expected* measures rather than Monte-Carlo scanpaths; trial-level dynamics are not modelled.

Disparate impact. The model is trained on corpus-mean RT, which advantages average readers. Per-reader bias is correlated with reading speed: slow readers receive systematically less accurate predictions than fast readers, even though the cascade structure can in principle account for them via per-group cog scalars. Deployment in reading-difficulty diagnostic settings would require per-reader calibration to avoid systematically misdiagnosing slower readers.

Comparison to related work. Hahn and Keller (2016) (NEAT) also train end-to-end against eye-tracking, but learn an opaque attention mechanism rather than preserving named cascade stages. SWIFT (Engbert et al., 2005) and OB1-Reader (Snell et al., 2018) are interpretable but hand-tuned. To our knowledge this is the first model that retains the named stages of Reichle et al. (2003) while being trained by gradient descent against raw text.

Conclusion

We presented the first end-to-end-trainable instantiation of an E-Z Reader-style cognitive cascade with a transformer LM backbone. The model is competitive with strong baselines on word-level fixation-time prediction, supports a double-dissociated dualctx specialization, and—most strikingly—recovers from optimization a textbook E-Z Reader prediction that fast and slow readers differ in lexical-stage parameters but not in motor or skip-decision parameters. Future work should bootstrap the per-group fit to confirm the cog dissociation with uncertainty bounds, parameterize L_2 with a semantic-integration signal, and extend the cascade to clinical reading populations (e.g. dyslexia), where parameter-localized accounts of reading deficits are scientifically and societally valuable.

Acknowledgments

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CRedit contribution statement

Efe Kahraman: Conceptualization; Data curation; Formal analysis; Investigation; Methodology; Software; Validation; Visualization; Writing – original draft; Writing – review & editing; Project administration.

Arturo Dalbagno: Group member.

Lorenzo Cipriani: Group member.

Meda: Group member.

Code and Data Statement

GECO (Cop et al., 2017) and Provo (Luke & Christianson, 2018) are publicly available eye-tracking corpora used under their original licenses. All training code, evaluation scripts, trained checkpoints, and analysis notebooks are released at

<https://github.com/E-8814/Neuro-EzR> (under preparation; full release on submission). No new human data were collected for this study; ethical approval was therefore not required.

Statement on the Use of Technology

Stage	Tools used & how
Prewriting / brainstorming	ChatGPT and Claude were used for ideation, structuring the paper outline, and clarifying argument structure.
Paraphrasing / rephrasing	Claude, ChatGPT, and Grammarly were used to rephrase sentences for concision and academic register.
Spelling / grammar checking	Grammarly was used throughout for spelling and grammar correction.
Formatting / references	References were formatted using BibTeX with the apacite style; no other automated reference manager was used.
Generating part of the text	The initial draft was substantially generated by Claude (Anthropic) on the basis of our project repository, then edited and revised by the authors. ChatGPT (OpenAI) was used for additional generation and revision passes.
Other generative-AI use	Claude Code (Anthropic) was used as a coding assistant during development of the cascade implementation, training pipeline, and analysis scripts.

Confirmation Statement. By submitting this assignment, we, Efe Kahraman, Arturo Dalbagno, Lorenzo Cipriani, and Meda, confirm that: the assignment is our own intellectual property and that ideas as well as language from other sources have been properly cited, referenced, and quoted (when necessary). All language and ideas not our own have been properly identified as such. We have disclosed all technology that we used in the writing process. We claim responsibility for our submitted work.